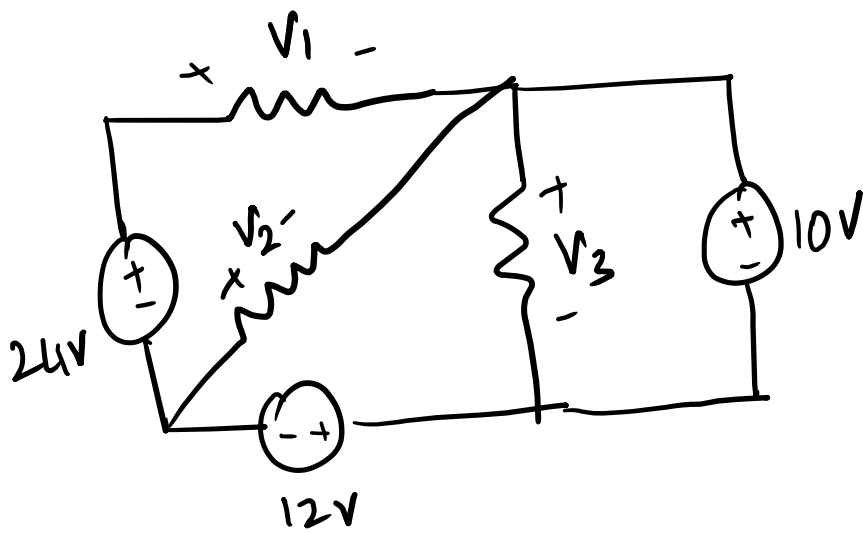




Find V_1 to V_3
using KVL

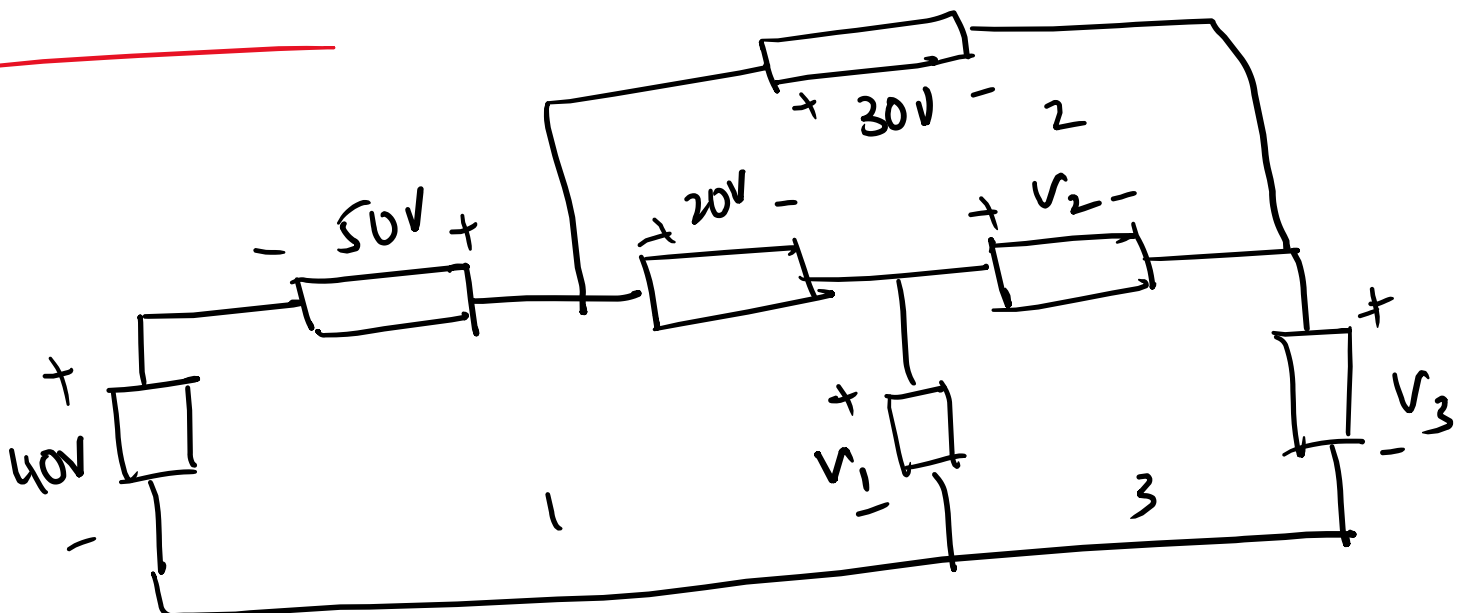
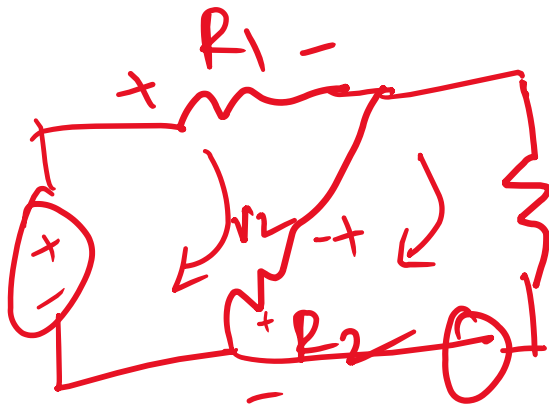
$$V_3 = \underline{10V}$$

$12 - V_2 - V_3$ loop : $-12 - V_2 - V_3 = 0$

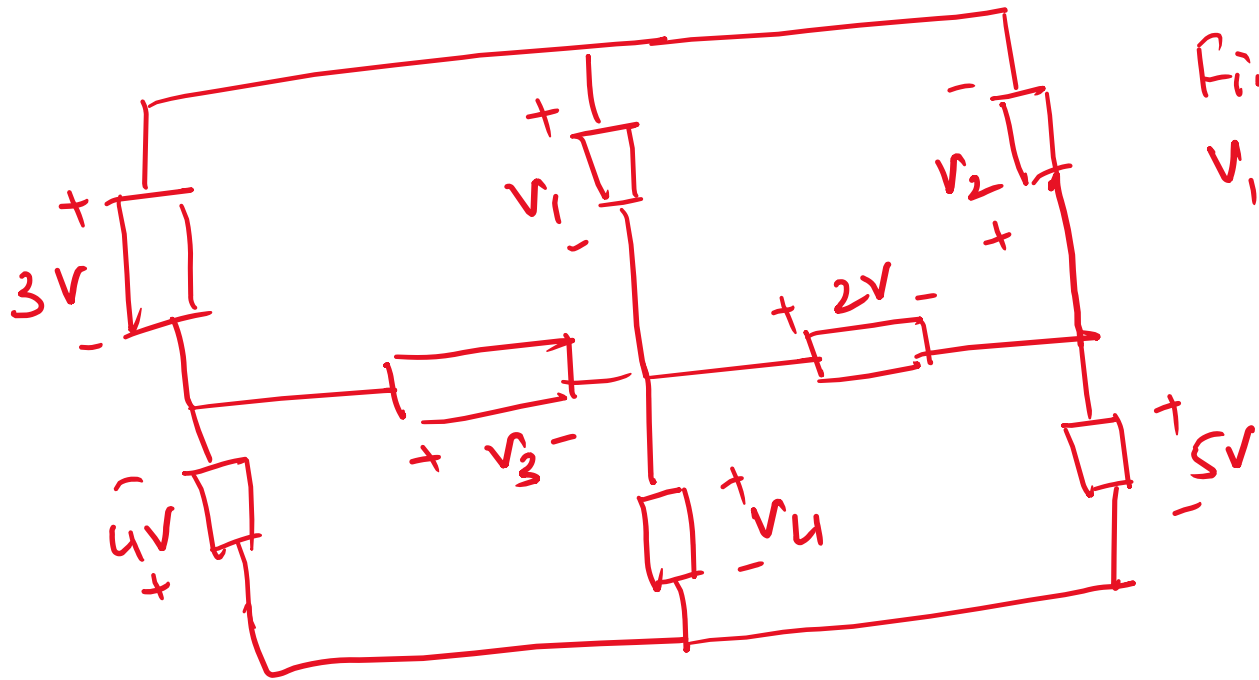
$$V_2 = \underline{-22V}$$

$24 - V_1 - V_2$ loop : $24 - V_1 + V_2 = 0 \rightarrow V_1 = \underline{2V}$

$24 - V_1 - 10 - 12 = 0 \rightarrow V_1 = \underline{2V}$



$$\begin{aligned}
 \text{loop 1: } 40 + 50 - 20 - V_1 &= 0 \rightarrow V_1 = 70 \text{ V} \\
 \text{loop 2: } 20 - 30 + V_2 &= 0 \rightarrow V_2 = 10 \text{ V} \\
 \text{loop 3: } V_1 - V_2 - V_3 &= 0 \rightarrow V_3 = 60 \text{ V}
 \end{aligned}$$



Find
\$V_1, V_2, V_3, V_4\$

$$\underline{V_1 = -8 \text{ V}} ; \underline{V_2 = 6 \text{ V}} ; \underline{V_3 = -11 \text{ V}} ; \underline{V_4 = 7 \text{ V}}$$

Nodal and Mesh Analysis

Nodal $\rightarrow$ Apply KCL to find node voltages.

Mesh $\rightarrow$ Apply KVL to find mesh currents

Nodal: • Node voltages are the circuit variables.

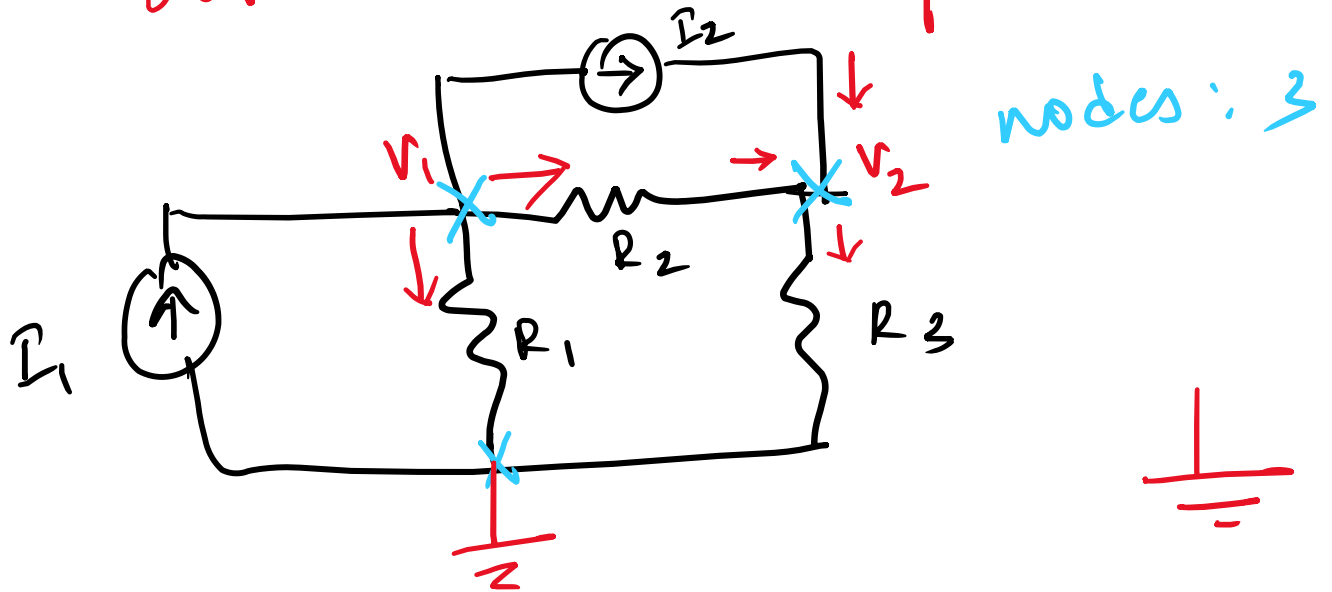
Given a circuit with ' n ' nodes

1. Select a reference node. Assign node voltages ($V_1, V_2, \dots, V_{n-1}$) to all remaining $(n-1)$ nodes.

2. Apply KCL to each of the $(n-1)$ non-reference nodes.

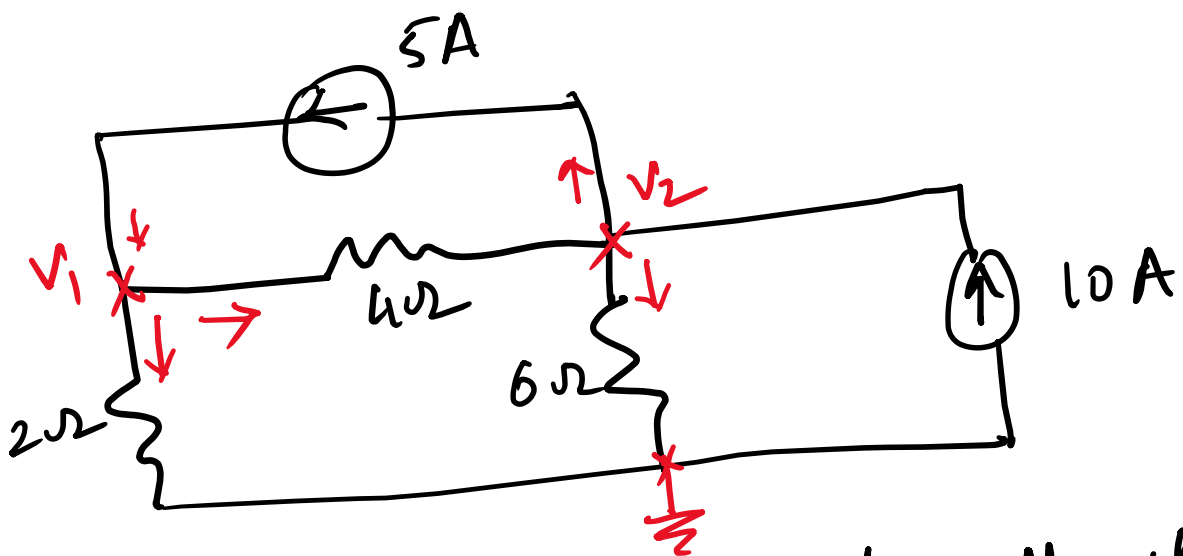
Use Ohm's law to express branch currents in terms of node voltages.

3. Solve simultaneous eqn's to obtain node voltages.



KCL @ node 1 : $I_1 = I_2 + \frac{V_1 - 0}{R_1} + \frac{V_1 - V_2}{R_2}$

KCL @ node 2 : $I_2 + \frac{V_1 - V_2}{R_2} = \frac{V_2 - 0}{R_3}$



Find the current through all the resistors using nodal analysis.

@ node 1 : $5 = \frac{V_1}{2} + \frac{V_1 - V_2}{4}$

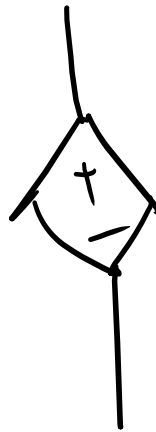
@ node 2 : $10 + \frac{V_1 - V_2}{4} = 5 + \frac{V_2}{6}$

$V_1 = 13.33V$; $V_2 = 20V$

Sources :



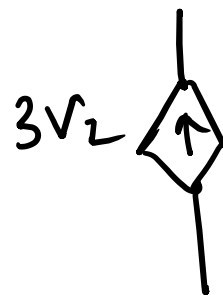
voltage
Independent
sources

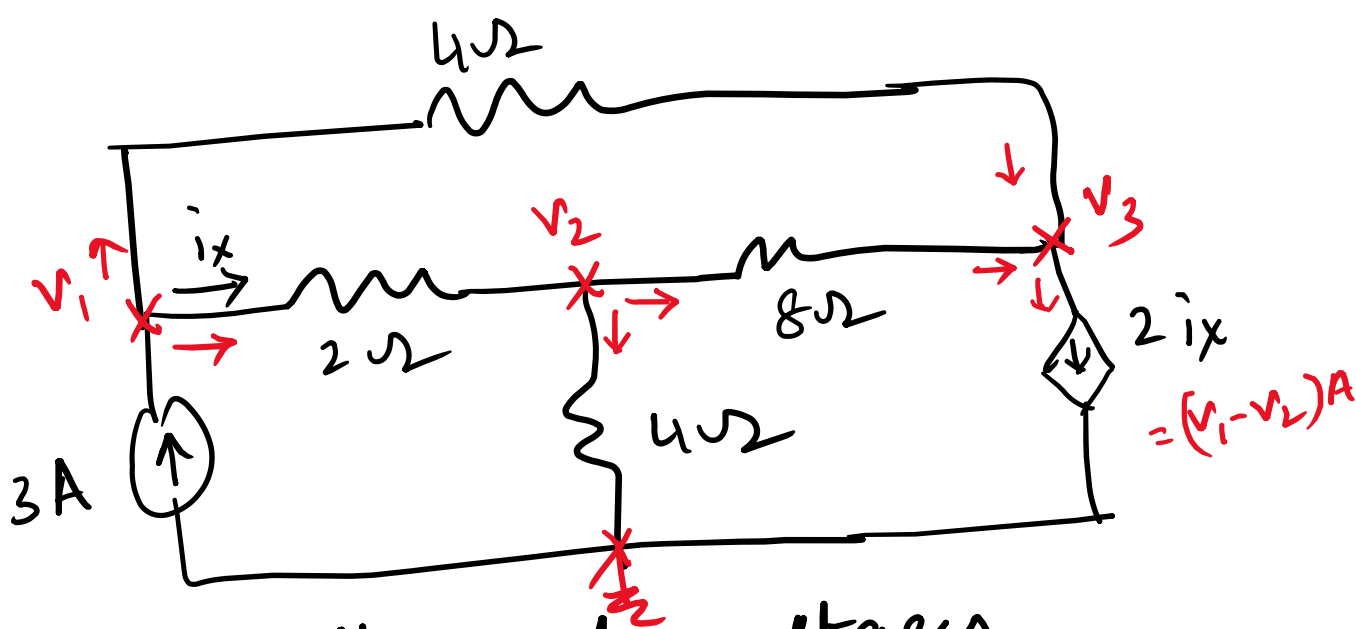


Dependent
sources.

Dependent source

Current : VC CS
CC CS





Find all node voltages.

$$i_x = \frac{V_1 - V_2}{2}$$

At node 1 : $3 = \frac{V_1 - V_2}{2} + \frac{V_1 - V_3}{4}$

At node 2 : $\frac{V_1 - V_2}{2} = \frac{V_2}{4} + \frac{V_2 - V_3}{8}$

At node 3 : $\frac{V_1 - V_3}{4} + \frac{V_2 - V_3}{8} = 2i_x = (V_1 - V_2)$

$$\begin{aligned} V_1 &= 4.8 \text{ V} \\ V_2 &= 2.4 \text{ V} \\ V_3 &= -2.4 \text{ V} \end{aligned}$$